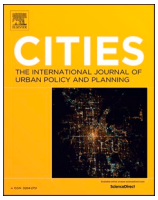
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Co-designing the urban energy transition: A resident-based approach

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ABSTRACT

The high ambitions for renewable carbon technologies, increasing electricity demands and technological developments will likely open for more small-scale electricity production closer to end users in cities and urbanized areas. This paper seeks to understand how participatory design methods and the integration of renewable energy techniques like photovoltaics can contribute to more democratic, sustainable, and livable urban planning in the age of energy transition. Data was analysed through participatory design workshops in a suburban district in the north of Sweden, with residents living and working there, and with stakeholders from the municipality and the local energy company. The results of the resident design workshop indicate a significant reservoir of knowledge and creative ability among the participants. It emphasizes the importance of bringing people together to share their opinions and ideas for a more democratic planning in the built environment. The stakeholder workshop indicates that co-operations between multiple stakeholders and the development of participatory methods are essential for addressing the problematic issues associated with e.g., building regulations and community shared electricity. Fostering collaboration, and promoting inclusive decision-making, can lead to more effective and sustainable practices, better regulatory frameworks, and improved outcomes for communities.

1. Introduction

Countries around the world have ambitious goals for renewable energy. Increased reliance on renewable energy is considered necessary to reduce the emissions of greenhouse gases and mitigate climate change, but also for concerns about security of supply (Vassiliades et al., 2023). In Europe for example, the Swedish government has set the goal that the country should be a net zero carbon society by 2045, at the latest, and that the energy sector should generate 100 % of its output from renewable sources (wind, solar, etc.) by 2040. Although the share of renewable electricity has increased substantially the last decade, large-scale hydropower and nuclear still dominate; these sources correspond to about 80 % of electricity supply (Swedish Energy Agency, 2019). The Swedish electricity system is thus highly centralized and dominated by large-scale production.

The goals for renewables together with the rapid development of renewable electricity technologies opens both for more small-scale electricity production and for increased participation of individual households as *prosumers* (**pro**ducers and con**sumers** of electricity) (Rizzo et al., 2020). To realize this development Sweden has established economic subsidies e.g., for installment of "grön teknik" (green technology) as photovoltaics (PV) for property owners to support the energy transition (The Swedish Tax Agency, n.d.).

Prosumers can play a key role in bringing zero-carbon energy production closer to end users (Rizzo et al., 2020). This has the potential to alter the structure of the Swedish electricity market by promoting more small-scale production and reducing centralization, especially since most end users live in urban areas and cities (Population Reference Bureau, 2016). Such production is also possible in more complex environments (Spett et al., 2024). Acceptance of renewable energy is crucial for developments in, or close to, urban and residential areas (Rizzo, 2017). On the one hand, developments in areas where people live may aggravate public acceptance while, on the other hand, increased participation, and engagement from residents could potentially build institutional capacity and learning. Previous researchers have found a positive relationship between general attitudes towards (yet mainly large-scale) renewable energy and acceptance (Dimitropoulos & Kontoleon, 2009; Ek & Matti, 2015; Liebe & Dobers, 2019), but it should be remembered that attitudes towards and acceptance of renewable energy is highly dependent on the local context.

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Although involvement of the local population is important, public acceptance of large-scale renewable energy developments has so far mostly been related to a relatively passive form of acceptance (consent). For property based small-scale solar power, individuals' active willingness to invest in and install the technologies is required. Property based small-scale solar power is however likely also motivated by expectations of personal benefits. Expectations about cost savings and other monetary benefits have been found important drivers for property based solar panel investments (Hille et al., 2018; Lee et al., 2018) but also knowledge, environmental motives and technology interest have been pointed out as important drivers (Cho et al., 2017; Stigka et al., 2014; Zhai & Williams, 2012).

Active participation can also be achieved by other means; from the perspective of urban and landscape studies, research into acceptance issues has deployed participatory design methods to engage users in the planning of future energy projects (Katsou et al., 2020; Rizzo et al., 2020). *Participatory Design* is a field that strives to include end-users as full participants of the traditional design and planning process (Mazé, 2007). Like transdisciplinary methods for architecture and urban planning (Rizzo & Galanakis, 2015; Tejedor et al., 2018), mock-ups, games, and temporary interventions are simple strategies to discuss ideas, regardless of technical and language skills.

The relationship between public acceptance and large-scale renewable energy establishments has been extensively researched while studies on small-scale electricity developments in urban areas are scarcer (Hille et al., 2018; Lee et al., 2018), although the literature on the market penetration of renewable energy and the factors enhancing or impeding this development is extensive.

Insights on issues that may support local engagement when new projects are planned are essential for policy makers and public officials in their endeavor to design and implement effective and legitimate measures to reach the goal of Sweden as a net zero carbon society. Results should also be useful for the solar power industry and for community-based cooperatives and networks engaged in renewable energy developments.

The purpose of this paper is to test a participatory design method on how to encourage and activate residents in the urban planning processes by using a hypothetical small-scale photovoltaic scheme integrated in a suburban district in the north of Sweden. The aim is also to see how participatory design methods can help to democratize the planning of the urban energy transition in Sweden by including its citizens.

The main research question is: - How can the urban energy transition be more democratic and accepted by including citizens in the design and planning process? The sub questions are: -What incentives can activate citizens to play a greater role in the planning process and potentially become prosumers? And: - By what means can the citizens become the driving force for future energy development?

2. Theories for transition

In a historical view *participatory design* was originated in Scandinavia in the 1970s and 1980s for democratically empowering and fostering democracy in the workplace by allowing workers to determine the shape and scope of new technologies (Spinuzzi, 2005). Practice-based approaches have the potential to articulate knowledge and politics actively by producing it through design. It can thereby provide essential knowing, and other forms of thinking that are lacking in other disciplines (Mazé, 2019).

In "participatory design" the term *participatory* is argued to refer to ethics, politics, democracy, and empowerment. How stakeholders or users, developers and planners by participation cooperatively make or adjust systems, technologies and artefacts in ways which fit more appropriately to the needs of those who are going to use them. The term *design* is the 'how' of designing, i.e., a focus on the practice of design the nature of design activities, the need for providing means for people to be able to be involved, the need for respect for different voices (Bannon & Ehn, 2013).

Previous results (Tejedor et al., 2018; Törnroth et al., 2022) indicate the importance of deploying a level of design thinking in small-scale energy projects that can be understood and controlled by the inhabitants. These findings hold significance for the growing number of design thinking and urban living lab experiments focused on energy projects across Europe. (see among others: Finocchiaro et al., 2014; Evans et al., 2015; Delinchant et al., 2016; Rizzo et al., 2021). Therefore, we used a framework proposed by Sanders et al. (2010) to engage non-designers in Participatory Design (PD) activities.

There are examples of existing opportunities for citizens to influence in a local planning process. In Sweden, where our case study is located, the municipality has when a development or a change is planned in an area and before a new or an amendment of an existing "detaljplan" (local plan) is legalized, a commission to monitor the public interest, and to find acceptance from citizens by for instance letting them appeal to a proposed measures local plan (Plan- och bygglag, 2010) is compulsory for new or amended "detaljplan" (local plan). Thus, those appealing the process are typically property owners. In this way consultation processes become a matter between those who have (a property right) and those who have-not (such a right) (Arnstein, 1969). A more recent democratic innovation was added to the "Kommunallagen" who permits municipalities to allow persons registered in the municipality to submit "medborgarförslag" (citizen proposals). The aim is to increase citizens' political participation (Svensk författningssamling, 2017) . This recent innovation could be seen as a step forward in Arnstein (1969) ladder of participation, although it's not enough to achieve full participation.

According to Torrekulla (2005) increasing renewable developments in urban areas will likely call for a more holistic understanding of energy and built environment whereby people and their surrounding environment are understood as a unity rather than an arithmetic sum of people, infrastructures, and so forth.

In a participatory process it is necessary to involve several standpoints for a stronger objectivity because many people get affected by intended measures in a building environment. According to Harding (1993) strong objectivity with culturewide beliefs also requires that we think of a strong reflexivity as it functions as evidence at every stage in a scientific inquiry (Harding, 1993). She claims that marginalized groups bring objectivity, opposite to most beliefs that the experts always know best in every situation.

Thus, even if it is not possible to involve all the citizens in a participatory process, since it's a matter of not only ownership but also of interest, still participatory methods can be beneficiary to gain a more democratic planning process by involving the citizens more than today, not only them that are directly affected, as the nearby property owners, but also the citizens living and working in an area.

Anthropological evidence suggests that human groups above 500 to 1500 people cannot coordinate to reach decisions that can set pressure on local governments (Alexander et al., 1977). Also, according to Alexander et al. people need an identifiable spatial unit to belong to, an identifiable neighborhood, that is distinct from all others. Therefore, architectural typological analysis can be a tool to identify neighborhoods in a district, where common production and consumption of electricity can become most suitable.

The willingness to invest in solar power is likely affected by expectations about direct personal benefits. Support for solar power is expected to be affected by attitudes in general, environmental orientation and perceptions about fairness and land use.

Prosumers are dependent on restrictions and regulations on monetary subsidiaries. The purpose of tax reduction for installment of green technology (for the cost of labor and materials), is to give incitements to private, and company real estate owners and institutions to produce small-scale renewable energy. It is also a question on the scale on the PV installation. The definition of what is defined as a small-scale energy facility, or what is not, can be seen from perspectives on regulations of the energy production, distribution, and taxation. According to the

"Legislation on taxation of energy" (Lag (1994)) producers of electricity distribution to a concession obligated grid is energy taxable. On one hand, so called, *micro producers* do not have to pay a grid fee for the input of electricity to the electricity grid and are entitled to a tax reduction from Swedish Tax Agency, and on the other hand, property owners are only allowed to construct facilities of a maximum of 500 kWh of solar electricity according to Lag (1994).

3. The case study

We used a residential area called Porsön, located in the municipality of Luleå in the northernmost part of Sweden, as case study. The characteristics of the area are both in a socio-demographic and in an architectural aspect not representative for a typical district in Sweden and will imply challenges regarding the outcomes of the workshops.

Thus, the motivation for selecting the area as a case study, despite some aproblematic circumstances, is that an innovative community-based renewable energy establishment could fit and enrich the development plan for "Porsö Centrum" (The Central Porsö District) that is presented later in this study.

3.1. Building typological analysis

In architecture, building typology can be described as identifying and grouping buildings and urban spaces according to the similarity of their essential characteristics (Pevsner, 1976). Typology is an essential concept in architecture, providing architects with a framework for understanding buildings and their roles in society by classifying buildings based on their function, form, and structure. (Caniggia & Maffei, 2001). Or as Simon Unwin puts it *"A place is established by a configuration of architectural elements that seems (in the mind informed by its senses) to accommodate, or offer the possibility of accommodation to, a person, an object, an activity, a mood, a spirit, a god."* (Unwin, 2009).

In our study this approach can be used as a tool to analyze in which places different solar cell applications should be integrated, but also to show that contextual differences play a crucial role for implementation of PV.

Diverse building typologies have been identified in the Porsö district and can be categorized into five main areas delimited for this study. The core areas: (A1), the university campus who forms its own enclave and commonly contains institutional buildings in a post-modern expressive industrial character (Fig. 3) and (C), the District Center, typical built like many other "ABC (Arbete, Bostad, Centrum) städer" (Work, resident, and center cities) during the Swedish "miljonprogram" (million program) 1965–1975, where the buildings are in a box like modernistic style. Both these building environments have a large-scale character (Fig. 2).

The residential areas (B) are divided into sub-areas and constitute high-raised apartment blocks (B1), high-raised student housing (B5, B6, B7, B8), low-raised apartment blocks (B2, B3), and single-family housing (B4).

There is grey infrastructure such as parking lots (P), car roads and bicycle and walking tracks adjacent to different facilities. There is a large amount of green infrastructure that surrounds the whole area, mainly made up by a large recreational park in the south, with open fields and groves of trees, and a gazebo and a small pond. (g) is an unused green field, and a former football pitch. (S) contains a gym and an outdoor football pitch. There is also a jail (J) nearby, isolated north of



Fig. 1. Processed satellite map with a typological analysis on the delimited study area of Porsön. Source: Satellite map from Google Earth.

the university area (Fig. 1).

3.2. Porsö development plan

In recent years, the area has experienced increasing problems with criminality like violence, vandalism, and fires. Therefore, a local development plan has been created to identify measures that can be implemented to increase the area's attractiveness and safety and foster feelings of inclusion and participation among the residents (Luleå kommun, 2024). The plan is part of a national initiative named "Framtidens goda stad" (Good cities of the future) lead by the Royal Swedish Academy of Science (IVA, 2017) and involving about 15 different stakeholders (including developers, inhabitants, students, university, etc.).

The development plan is a deepening of the municipality's "översiktsplan" (master plan). It will set out directions for the long-term development of the district and provide guidance for decisions e.g., how to develop and preserve the built environment. The development plan aims to contribute to: Greater equality between groups; Increase the active participation of residents in community life; Increase well-being and pride and increase the attractiveness of business establishments (Luleå kommun, 2024).

The project to upgrade the district center has started in 2022. It includes a thorough renovation of all land surfaces together with renovation of certain facades, new canopies, new lighting. The solar staircase (See Fig. 4) gives people according to "Luleå kommun" (Luleå kommun, 2024) reasons and opportunities to stop, sit down and meet other people.

4. Methodology

The framework we used developed by Sanders et al. (2010) was to engage non-designers in participatory design (PD) activities. It is described in three dimensions *form, purpose,* and *context*. *Form* is the activity taking place between the participants, i.e., making, telling and/or enacting. *Purpose* is the description of why the tools and techniques are being used. This is done in four dimensions, first to *probe* participants; secondly to *prime* the participants by immersing them for the domain of interest; thirdly to get a better understanding of their *experience* and lastly four to *generate ideas* or *design concepts* for the future.

This framework is used to decide which tools and techniques are most relevant for specific situations and can be used as a pedagogic device that helps to connect design and research, academics, and practitioners (Sanders et al., 2010).

The research plan was to first recruit participants from the focus group engaged in a previous survey about resident acceptance for solar power in Porsön. Initially, there were 35 people contacted on a list from this pre-study. After the recruitment an agenda for the workshop was sent out for the purpose of informing the attendees of the coming activities. A flyer was created to announce the event, with the idea of marketing the event and recruiting more participants, like residents, employees of the university, and students living and working in the area. The *form* of the workshop (here called WS1 – the workshop with the residents and WS2 – the workshop with the stakeholders). WS1 was a face-to-face meeting held in October 2021 in the architectural studio at Luleå university of technology where we conducted the workshop in one group with five residents. Before we started an oral consent was given to take pictures and audio record for documentation of the workshop.

The agenda for the WS1 was divided in three steps: first an informative part (2D collages), (Table 1 – making tangible things) secondly a creative part; this was, in turn, split in two exercises: a first one to select appropriate locations for PV installations (2D mapping) and a second one to design instalments on the chosen places (3D mockup).

The purpose of priming the participants by shortly briefing them about the research project and giving the delimitations of the study area and the intended outcomes of the workshop, was that they could easier come into a design thinking mode. After this initial information part, the creative part could then begin.

In the first step we started to work analogically with sketching paper, post IT patches and markers as tools to discuss the most suitable locations for implementation on a printed-out satellite map of the delimited area. All the participants were gathered round one table. In the background, on a projector screen, a variety of contemporary designs of different PV solutions on green and grey infrastructure and on buildings were envisioned (2D collages). The scenario was that new installations could be integrated in parks, squares, parking lots, and on the edifice of buildings, e.g., roofs, facades, window shutters and so forth. There was no limitation where it could be applied so our prerequisite was that it could be either in public places or/and on buildings. Additionally, a solar map of Luleå municipality was used, displaying the best solar gain conditions on roofs. The idea was that the participants should first locate which areas were most suitable and then compare them with the solar map. Also, 3D views were handed out to get greater empathy for the area. We also had the opportunity to search google earth for additional views and street views. After the first creative part on location a presentation was held by one of the participated residents followed by a common discussion. The presentation and discussion were audio recorded for later transcription and analysis of the result (Fig. 5).

In the second step of the creative part the 3D mock-up part, (Fig. 6) the participants used model materials i.e., foam, folio, cardboards, plasteline, wooden plates, rods and so forth. A demonstration of how to use the various materials to build models was shown by the facilitators. To give inspirational ideas, a slide show running in the background on the projector screen showed inspirational images, e.g., canopies with integrated PV for cars, consistent with parking lots that are present in the university area. The ideas came easier when we gave keywords and used words like playfulness, vertical solar panels, installations in open spaces.

A following workshop (WS2) with stakeholders, i.e., the city architect from Luleå municipality, and a representant from the local energy

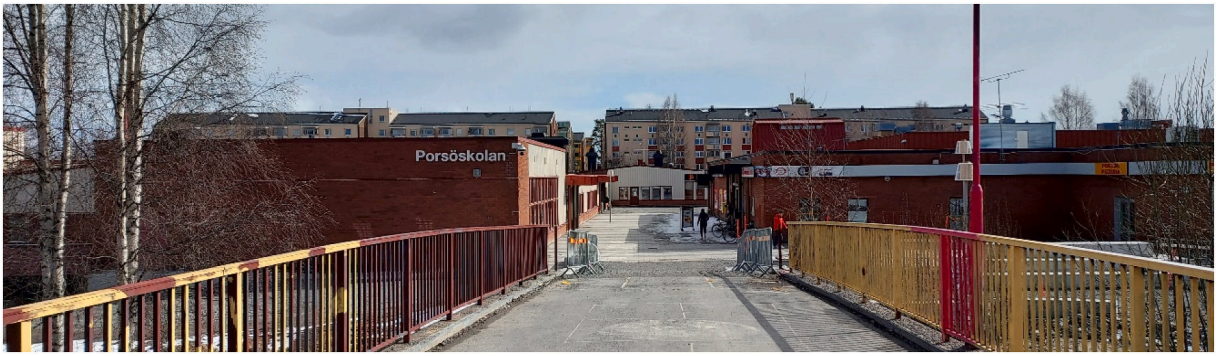


Fig. 2. View from "Kunskapens bro" (bridge of knowledge) towards Porsö District Center (C).



Fig. 3. View from Porsö District Center towards the campus area of Luleå university of technology (A1).



Fig. 4. A new public space with a "solar" staircase is under construction. It is connected to "Kunskapens bro" (The bridge of knowledge) that runs between Porsö center, and the university campus. Source: WSP.

company "Luleå Energi", consisted of a face-to-face meeting at Luleå university of technology in November 2021, were we informed about the outcome of the resident workshop and had a discussion around the following questions:

"1. Future collaborations on solar energy?" "2. What are the technical limitations?" "3. What is possible to do and what are the obstacles?"

The approach was not so much a participatory design workshop but more a deliberative discussion, with answers and questions by and for the participants. We worked on mural boards with each question on three boards. As tools, we used post-IT patches on which the participants could write down their answers and ideas. The patches were then mounted on a board categorized by the three main questions. The workshop was an open discussion, as we asked more questions around each main question. Documentation was done by note taking and transcription.

5. Results

5.1. The resident design workshop

The results were collected through transcriptions of audio recordings after each presentation part. One person from the resident group presented and the other participants filled in with comments or came with additional ideas, also the facilitators asked questions to encourage the discussion. The presentation part thereby became an extended part of the discussion the resident has had during the creative parts.

In the beginning the participants' focus was mainly directed on political, technical, and economic issues related to solar integration, but not so much on soft values and on aesthetics and architectural design. This circumstance was observed in the participants' discussion during the event. This could have been caused by the fact that the participants were selected among the respondents to a survey distributed some months before our workshop. This former survey was investigating factors potentially affecting willingness to invest in solar power.

In the "location" exercise, there were some alternative places and measures discussed in the beginning of the presentation (Fig. 1). According to the residents' suggestion, the most suitable places for PV integrated installations could be the apartment block area (B2), particularly in the south part of the area on buildings with roofs facing southwest. This could be more feasible in buildings organized as condominiums, where each apartment owner has more influence and can gain more from solar installations. Integration could be for instance on garage structures, carports and parking spaces. The campus area (A1), with considerable amount of space on low inclined roofs, was also

Table 1
A refined and adapted framework became the methodology for the workshops. A table for the tools and techniques of participatory design organized by form and purpose. WS1 was the resident design workshop. WS2 was the stakeholder workshop with representatives from the municipality and the local energy company.

Tools and techniques	Probe	Prime	Understand	Generate	WS 1	WS 2
MAKING TANGIBLE THINGS						
2D-COLLAGES using visual and verbal triggers	X	X	X	X	X	
2D-MAPPING using visual and verbal components		X	X	X	X	
3D-MOCK-UPS using e.g., foam, clay, wood model material, folio			X	X	X	
TALKING, TELLING AND EXPLAINING						
Post-IT patches to organize, categorize and prioritize ideas.			X	X		X
ACTING, ENACTING AND PLAYING						
PARTICIPATORY ENVISIONING AND ENACTMENT by setting users in future situations				X	X	X
PRESENTING			X	X	X	

mentioned as an alternative placement. There, PV could even be used for promoting the positioning of the university at the forefront in the developing of cutting edge renewable technologies. Other solutions were to have walkable pavements in the campus area, for example on the "Regnbågsallén" (Rainbow Alley), a walking alley that is the campus main walkway.

In the end, the vast parking spaces around the campus (Fig. 1, and the symbol P for (car parking) were chosen as the first place for working with the integration of PV in the model making part of the workshop. On that area solar roofs could be used as charging points for electric cars, and engine heaters powered by solar energy. (Fig. 6, right) The second chosen place was a green park area (g), close to a former football pitch, no longer in use, which could be converted into a playground for children. The participants created here a stand-alone structure that could follow the movements of the sun, a "solar sculpture" not too fragile nor dangerous for children playing around. This "sculpture" could also be used functionally as a sitting ground with benches under movable fundaments. Such a structure could also be used by children for a pedagogic matter to learn and get acquaintance to the new technology (Fig. 6, Left).

The results of the design workshops indicate a significant reservoir of knowledge and creative ability among the participants. It emphasizes the importance of bringing people together to share their opinions and ideas for democratic planning in the built environment.

Results also give indications that not only that energy issues can be solved but also an improvement of the built environment can be achieved in the energy transition. Positive synergetic effects on the building environment can thus be created.

5.2. *The stakeholder workshop*

Through the stakeholders we explored existing restrictions and regulations that could affect the implementation of small-scale photovoltaic schemes. The result of the stakeholder workshop showed that the issue of ownership is an obstacle. In Sweden, property owners are allowed to



Fig. 5. Left. Resident location part. Right. Their suggestions of possible locations with notes on post-IT patches on the satellite map.

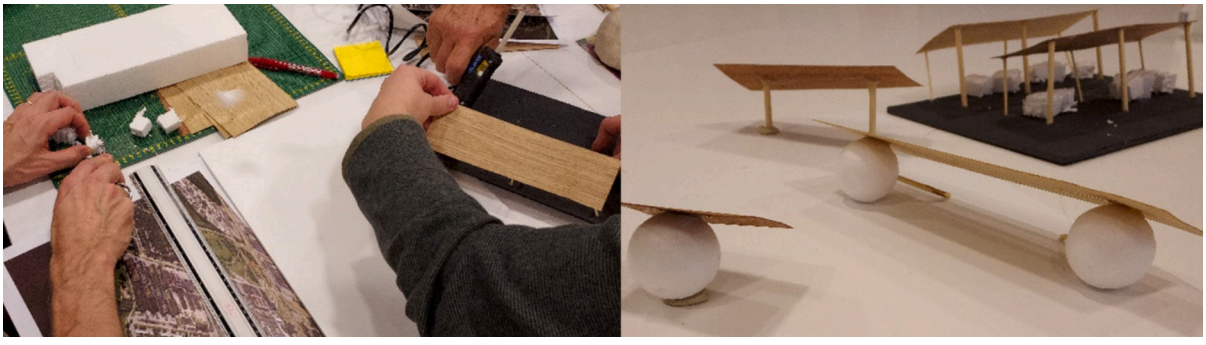


Fig. 6. Left. Resident modelmaking. Right. The outcome of the modelmaking part. A stand-alone sun-following solar (in the forefront) and structure and the solar carparking canopies (in the back).

construct facilities of a maximum of 500 kWh of solar electricity because of legislation on taxation of energy (Lag (1994)). This is problematic, for instance, in case different property owners want to create an energy community by placing a large photovoltaic installation on the property with the best exposed roofs and sharing the energy there produced among the other properties. There is also an issue with tax declaration, since the owner who has the PV installed on the roof would have to pay all the taxes.

The construction of flat roofs is another obstacle for implementing building applied photovoltaics (BAPV). In the campus area with a lot of low inclined roofs, a stable roof is needed to carry the load of solar cells. On older buildings, new installations of building applied solar panels can require a reinforcement of the roof structure, or a totally new roof construction. Maintenance work with extra weight for the roof construction when snow shoveling during wintertime must also be considered.

The issue of lack of effective collaboration can be an obstacle i.e., "Luleå Kommun" (Luleå Municipality) has collaborated with "Luleå Energi" (Luleå Energy - the local energy company) on solar integration. In a common idea for PV installations in a roundabout there were obstacles with regulation on traffic security and issues of vandalism.

6. Discussion

The purpose of the design workshop was to test a participatory method and later analyze whether the suggestions and input from residents could enhance democratic planning in the built environment. By involving residents in the decision-making process and incorporating their perspectives, it aimed to promote a more inclusive and people-centric approach to urban planning.

The main research question was: - How can the urban energy transition achieve more acceptance and be more democratic by including citizens in the design and planning process? Based on our results we can argue that several objectives can be achieved by including residents in design and planning projects:

Firstly, it can empower the residents living and working in the areas for the planned transitions. Even thus the participants were few (only 5 people), we could observe the positive impact such an approach could have for future design and planning processes. The results also acknowledged that the participating residents had valuable knowledge about their neighborhood and therefore it should be an advantage for the whole community to include residents in the decision-making e.g., as they have responsibilities to what directly impacts their living environments.

Secondly, local knowledge and insights from design workshops can inform planning decisions and enhance democratic planning. By incorporating residents' suggestions (as in WS1) with the local stakeholder representative's knowledge (as in WS2), the results indicated that an improved form of participatory design is needed. The results from the stakeholder workshop also indicate that co-operation between stakeholders and the development of participatory methods are essential for addressing the problematic issues associated with e.g., building construction and regulations. Our assumption is that this improvement can lead to a better decision-making process and, ultimately, to more effective and responsive urban planning outcomes.

The first sub question was: -What incentives can activate citizens to play a bigger role in the planning process so they in the end can become prosumers easier? Based on our results we can argue that monetary subsidies can play a crucial role. Today the tax subsidiaries are directed to individual property owners, but they should instead be directed to a district or a neighborhood scale. Thereby, more citizens can be involved in planning for the energy transition, not only the owners but even the "have-nots". With this approach, more knowledge can come into the planning, and more affected people can be included. Even though not all can participate in urban energy planning processes, as it is a matter of interest, they can indirectly get benefits from it. e.g., monetary benefits for those that live in the same neighborhoods where the transition will happen.

The second sub question was: - By what means can the citizens own the question for small-scale renewable electricity production in their neighborhood and become the driving force for the future energy development? Based on our results we can argue that if the residents are included in the process from the beginning, they will have a sense of "owning the question" when they are part of a deciding process. Their acceptance will thereby inevitably increase.

One intriguing observation from the results is that the residents chose public spaces rather than actual buildings for integration. The preference for public spaces over buildings in the integration process highlights the value placed on community interaction, inclusivity, and holistic urban planning. Other reasons for this preference can be the character of the solar panels today i.e., a squared static form applied or integrated on e.g., roofs and facades with less room for creative design. Also, the issues of ownership can have influenced the decision. This result must face the issues of vandalism but could be prevented with a contextual design rooted and initiated by and for local inhabitants.

After a residential participatory stage, and standpoints from local stakeholders, it is necessary to further develop the ideas and standpoints into concrete proposals. In this process, involving experts becomes crucial, as they can bring objectivity by considering multiple subjective standpoints and incorporating their expertise. The experts can be actors with special knowledge about i.e., architecture, landscape architecture, building conservation, building construction and PV product developing. This idea and proposals are an iterative process between the residents, the experts, and the authorities for appraisal, analysis, and synthesis.

7. Conclusions

Empowering inhabitants through design thinking methodologies has several benefits. Firstly, it ensures that the perspectives and needs of the community are integrated into the design process. This can lead to more contextually appropriate and responsive solutions that reflect the aspirations and values of the residents.

Secondly, by involving inhabitants in the design process we can foster learning opportunities and facilitate the formation of new networks within the community. By actively participating in the design and decision-making process, residents can gain knowledge about energy systems, sustainable practices, and the built environment.

Also, there is a need to consider diversities in climate, architecture and culture when designing participatory processes. Starting and developing the participatory process at an intermediate, district or neighborhood scale, rather than at a large regional scale or single-family house scale, can help address these diverse contexts more effectively. Additionally, it is essential for the process to be municipality-driven and facilitated to account for the specific local contexts.

The reasons why an intermediate or neighborhood scale is beneficial for participatory processes are, among others: 1) Contextual Relevance: By focusing on an intermediate or neighborhood scale, the participatory process can directly address the specific climate, architectural styles, cultural practices, and political dynamics of the area. This allows for more contextually relevant discussions, solutions, and decision-making. 2) Proximity and Connectivity: Operating at a neighborhood scale enables closer proximity and stronger connectivity among participants. This facilitates more meaningful and inclusive interactions, as residents can directly engage with their neighbors and collaborate on shared concerns and aspirations. It fosters a sense of community and encourages active participation. 3) Tangible Impact: Participatory processes at an intermediate, district or neighborhood scale have the potential to bring about tangible and visible changes within a community. The outcomes of the process, such as the development of community spaces or improvements in infrastructure can directly benefit the residents and enhance their quality of life. 4) Knowledge Sharing and Learning: By

working at a localized scale, participants can share their local knowledge, experiences, and expertise with one another. This knowledge sharing fosters mutual learning and helps build a collective understanding of the challenges and opportunities specific to the neighborhood. It also allows for the exchange of best practices and innovative ideas relevant to the local context.

By involving residents in the planning process, trust between residents and planners can be built and community engagement can be strengthened. This collaborative approach can bridge the gap between experts and the community, promoting a more inclusive and transparent decision-making process.

By adopting design thinking methods that are accessible to all affected stakeholders, including inhabitants, power dynamics can be rebalanced, ensuring a more inclusive and equitable decision-making process.

Regarding the municipality-driven approach, involving the local government as a key driver of the participatory process is essential for several reasons: 1) Authority and Resources: Municipalities have the authority and resources to initiate, facilitate, and support participatory processes effectively. They can allocate funding, provide technical expertise, and coordinate various stakeholders to ensure the success of the process. 2) Local Governance: As the governing body at the local level, municipalities have a deep understanding of the political, social, and economic dynamics within their jurisdiction. Their involvement ensures that the participatory process aligns with local governance structures and decision-making processes. 3) Representation and Accountability: Municipalities are responsible for representing their constituents' interests. By driving the participatory process, they can ensure that a broad range of voices are heard and that the outcomes reflect the diverse needs and aspirations of the community. They are also accountable to the residents, providing a mechanism for transparency and accountability in decision-making.

Property owners may often drive the initiation of PV integration based on subsidiary and monetary benefits, but it is important to adopt inclusive approaches that engage a broader range of stakeholders. By promoting community-based approaches, providing incentives and support, raising awareness, and ensuring supportive policy frameworks, the integration of PV systems can become more inclusive, collaborative, and beneficial for the entire community.

Finally, the key recommendation to urban-energy policy makers is that action-oriented, transdisciplinary approaches to co-produce energy projects can be a way forward to include different segments of prosumers in future, small-scale urban energy projects.

Disclosure statement

During preparation of this work the authors used ChatGPT / chat.openai.com to get help with expressions in the English language in some part of the text. After using this tool, the authors reviewed and edited the content as needed and they therefore take full responsibility for the content of the publication.

CRediT authorship contribution statement

Lars Vikström: Conceptualization, Data curation, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Kristina Ek:** Writing – original draft. **Andrea Luciani:** Investigation, Methodology, Supervision, Writing – review & editing. **Agatino Rizzo:** Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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